

# Syllabus

**MAT 4233 — Modern Abstract Algebra** Fall 2026 · UT San Antonio

## Texts and Materials

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### Primary text

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*. Ch. 3 (complex numbers, polar form, roots of unity) and Ch. 13 (Möbius transformations).

### Supplementary reading

- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*. Not held by UTSA. Ch. 3 distributed as a handout; a copy is on reserve.

### Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*. Survey reading for the final lecture: statements and examples; proofs optional.
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*. Notices of the AMS 55 (2008), no. 10. Co-reading for the Möbius lecture.

## Course Information

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### Catalog Entry

MAT 4233 Modern Abstract Algebra

Basic properties and examples of semigroups, monoids, and groups, detailed study of permutation, dihedral, and congruence groups, cyclic groups, normal subgroups, quotient groups, homomorphism, isomorphism theorems, direct products of groups, rings and fields and their basic properties, ideals, polynomial rings.

### Prerequisites

MAT 3233 Modern Algebra or equivalent

### Credit Hours

3

### Course Modality

Traditional in-person

### Class Meetings

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|          |                         |
|----------|-------------------------|
| Duration | Wed 19 Aug – Thu 3 Dec  |
| Campus   | Main Campus             |
| Location | MS 2.02.12              |
| Time     | Tue Thu 2:30 PM–3:45 PM |

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## Learning Objectives

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By the end of this course you should be able to:

- Define the notions of group, cyclic group, subgroup, normal subgroup, homomorphism, isomorphism.
- Define symmetric and permutation groups; determine the cycle decomposition and parity of finite permutations.
- Define the notions of ring, subring, domain, field, ideal, homomorphism, isomorphism.
- Define rings of polynomials over a field, the notion of degree, and irreducible factors.
- Explain the abstract algebraic properties of various rings and number systems including  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{H}$ , modular rings  $\mathbb{Z}/n\mathbb{Z}$ , finite fields  $\mathbb{F}_p$  and polynomial rings  $\mathbb{F}[x]$ .
- Explain the common and specific aspects of the division and Euclidean algorithms in the rings  $\mathbb{Z}$  of integers,  $\mathbb{Z}[i]$  of Gaussian integers, and polynomial rings  $\mathbb{F}[x]$ .
- Show that every Euclidean domain is a Unique Factorization Domain.
- Use properties of Gaussian primes and factorization in  $\mathbb{Z}[i]$  to characterize sums of pairs of perfect squares.
- Communicate concepts in writing and debate technical arguments orally.
- Use a computer algebra system to for numerical and visual explorations in algebraic structures (e.g., the RSA cryptosystem and geometric transformation groups).

## About the Instructor

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Eduardo Dueñez, Associate Professor of Mathematics.

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|---------------|-------------------------------------------------------------------------|
| Department    | Mathematics                                                             |
| Office        | FLN 4.01.11                                                             |
| Student hours | TBD                                                                     |
| Email         | <a href="mailto:eduardo.duenhez@utsa.edu">eduardo.duenhez@utsa.edu</a>  |
| Homepage      | <a href="https://supernumero.us/about">https://supernumero.us/about</a> |

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Email is the preferred method of communication.

## Assessment and Grading

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### Weight of course activities

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| Component      | Weight |                     |
|----------------|--------|---------------------|
| Q&A sessions   | 40%    | 5 sessions, 8% each |
| Problem sets   | 15%    | 5 sets, 3% each     |
| Midterm Exam 1 | 20%    | Thu 1 Oct in class  |
| Midterm Exam 2 | 25%    | Thu 3 Dec in class  |

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### Q&A sessions

Five class meetings throughout the semester (roughly every other Thursday except when adjacent to an exam) will be *Q&A sessions* in which each student will be asked to explain and defend their solution to a homework problem submitted as part of a written assignment. You will only be asked questions related to an assigned problem and about the solution you submitted.

Each Q&A session is scored 0–3:

- **3** Correct explanation to a correct solution, demonstrating full understanding of the concepts and all aspects of the solution submitted.
- **2** A completely correct solution submitted is explained to an incomplete degree; **Or**: an unfinished solution, with a thorough and articulate account of what you tried, explaining where and why you got stuck while demonstrating knowledge of the background and reading necessary to solve the problem.
- **1** Correct solution inadequately explained revealing little or no understanding; *or*: partial solution accompanied by an incomplete but partly correct oral explanation.
- **0** Nothing submitted; *or*: work submitted (even if fully correct) is backed by no cogent explanation nor knowledge of concepts involved.

**A correct answer you cannot explain is worth less than an explainable honest failure.**

**Note:** A Q&A average  $\leq 1.0$  automatically caps the course grade to the level of ‘D’ (at most).

### Problem sets

Five sets, each with three or four theoretical problems plus one SageMath computer exploration. **Maximum three pages.** Do *not* submit code; include the relevant snippets and output as part of your written submission. (SageMath problems are also included in Q&A.)

Each item will be scored *only* based on the **degree of completion** (not on correctness): **2** for a nominally complete solution, **1** for a clearly partial one, **0** for nothing submitted or for obvious filler with no substance.

*Note:* Any submission containing non-human-readable content, stray characters, or artifacts revealing non-proofread AI output scores 0 for that item.

### Use of Generative Artificial Intelligence

**AI assistance is allowed in preparing work for submission**, as are any solution keys, collaboration, and internet sources. You are ultimately responsible for your submissions; any part of your work that has clearly not been proofread (e.g., includes non-human readable characters or LLM artifacts) automatically earns a zero grade. *Include one line noting what resources you used in your submission.*

This policy is by design, and it reflects how the course is graded: 85% of the grade comes from Q&A sessions and in-class exams where only what is truly understood and can be explained counts. Assignments submitted but not written/understood by you result in failing grades in Q&A sessions and exams. Use AI to learn faster —not to prepare and turn in papers you cannot defend, blocking you from passing exams as well.

### Exams

Both exams are traditional in-class (written and closed book). Exams are based **primarily on the assigned problem sets** —expect some context and phrasing to differ from the homework, plus a minority of other unseen problems. Memorizing solutions will not result in good exam grades; preparing by understanding the assigned problems will make the exams straightforward and earn high Q&A grades as well.

### Attendance and missed work

No extensions on problem sets. A *single* missed Q&A session in the semester may be made up during office hours within seven days (no documentation is required). *No Q&A scores are dropped.*

### Final grade ranges

| Grade | Range       | Grade | Range      |
|-------|-------------|-------|------------|
| A     | [90%, 100%] | A-    | [85%, 90%) |
| B+    | [80%, 85%)  | B     | [76%, 80%) |
| B-    | [72%, 76%)  | C+    | [69%, 72%) |
| C     | [65%, 69%)  | C-    | [60%, 65%) |
| D     | [50%, 60%)  | F     | [0%, 50%)  |

## Time Commitment Expectations

Expect an average of about **9 hours per week**, and no fewer than 7.

- Attend class meetings (**3 hours per week**)
- Reading the assigned textbook sections (**2–3 hours per week**)
- Problem sets and the Sage exploration (**2–3 hours per week**)
- Office hours and study group (**1 hours per week**)

## Additional Course Information

Each student should be intimately familiar with the contents of any work submitted for grading. The instructor has the right to request adequate **verbal explanation** of methodology and content for any submitted work. Credit will not be awarded when such an explanation is requested but not satisfactorily provided.

## Course Schedule

| Date       | Activity       | Topic                                                              |
|------------|----------------|--------------------------------------------------------------------|
| Thu 20 Aug |                | Course launch; divisibility and the division algorithm             |
| Tue 25 Aug |                | Primes, the Fundamental Theorem of Arithmetic, gcd and lcm         |
| Thu 27 Aug |                | Semigroups, monoids, groups — all of them $\mathbb{Z}/n\mathbb{Z}$ |
| Tue 1 Sep  |                | Subgroups; the unit group $(\mathbb{Z}/n\mathbb{Z})^*$             |
| Thu 3 Sep  | <b>Q&amp;A</b> | Defense 1                                                          |
| Tue 8 Sep  |                | Group homomorphisms; the order of an element                       |
| Thu 10 Sep |                | Cyclic groups; primitive roots                                     |
| Tue 15 Sep |                | Cosets and Lagrange's theorem                                      |
| Thu 17 Sep | <b>Q&amp;A</b> | Defense 2                                                          |
| Tue 22 Sep |                | Normal subgroups and quotient groups                               |
| Thu 24 Sep |                | The homomorphism theorem                                           |
| Tue 29 Sep |                | Euler's theorem, fast modular exponentiation, and RSA end to end   |
| Thu 1 Oct  | <b>Exam</b>    | Midterm Exam 1 — Segment 1                                         |
| Tue 6 Oct  |                | Rings and subrings                                                 |
| Thu 8 Oct  |                | Ring homomorphisms                                                 |
| Thu 15 Oct |                | Ideals                                                             |
| Tue 20 Oct |                | Quotient rings                                                     |
| Thu 22 Oct | <b>Q&amp;A</b> | Defense 3                                                          |
| Tue 27 Oct |                | Fields and skew fields; $\mathbb{F}_p$                             |
| Thu 29 Oct |                | Integral domains and the field of fractions                        |

| Date       | Activity       | Topic                                                                                                                           |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| Tue 3 Nov  |                | Unique factorization: UFDs                                                                                                      |
| Thu 5 Nov  | <b>Q&amp;A</b> | Defense 4                                                                                                                       |
| Tue 10 Nov |                | PIDs and Euclidean domains                                                                                                      |
| Thu 12 Nov |                | Complex numbers: arithmetic, the plane, polar form, roots of unity                                                              |
| Tue 17 Nov |                | Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points |
| Thu 19 Nov | <b>Q&amp;A</b> | Defense 5                                                                                                                       |
| Tue 24 Nov |                | The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey                          |
| Tue 1 Dec  | <i>Review</i>  | Review — the exam pool                                                                                                          |
| Thu 3 Dec  | <b>Exam</b>    | Midterm Exam 2 — Segment 2 weighted                                                                                             |

Readings for each meeting and assignment due-dates are on the [Weekly Schedule](#) (also available as a [printable PDF](#)).

## Department and University Policies

### Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at [ilse.rosalesmartinez@utsa.edu](mailto:ilse.rosalesmartinez@utsa.edu).

### Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law.

### Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes or corrections to the course materials, assignment dates, or other updates will be communicated to students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

### Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) for resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support page](#).
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#), or call (210) 458-4157.

- Your well-being is important. Support, including 24/7 mental health services, can be accessed from the [Well-being at UT San Antonio](#) site.
- [Student Assistance Services](#) offers confidential support and personalized guidance.
- The [Roadrunner Pantry](#) provides free access to groceries, toiletries, and other essentials.
- Students are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Review [Section 203 of the Student Code of Conduct](#).
- Visit [UT San Antonio Libraries and Museums](#) for journals, research tutorials, and your department's librarian.